

DR-2 cross-scale induction: the decoupling-iteration energy bound for arbitrary finite sphere sets

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

first issued 2026-06-08 · this version issued 2026-06-08 · v1.0

1. Purpose and scope

dr2-decoupling-closure v1.1 proved DR-2 for δ -separated $Q \subset S^2$ (T6 conditional on decoupling) and left the ARBITRARY- Q (non-separated) case at T4, with the cross-scale energy summation as the named residual. This note writes that step: the decoupling-iteration inequality and the induction-on-scales. It is conditional on the same decoupling black box; it does not reprove decoupling, and it cites (does not reproduce) the tight multi-scale constant bookkeeping.

2. The decoupling-iteration inequality

Lemma (iteration). Let $Q \subset S^2$ be finite, $\delta > 0$, and $\{\theta\}$ the partition of the sphere into $\delta^{1/2}$ -caps; write $Q_\theta = Q \cap \theta$. Then, conditional on ℓ^2 -decoupling at $p = 4$,

$$E_+(Q) \lesssim_\epsilon \delta^{-\epsilon} \left(\sum_\theta E_+(Q_\theta)^{1/2} \right)^2.$$

Derivation. By the majorant of v1.1 §2.1, $E_+(Q) \lesssim \int |f_Q|^4 \eta_R$, $R \sim \delta^{-1}$, $f_Q = \sum_{q \in Q} e(q \cdot x)$, whose Fourier support lies on $Q \subset S^2$. Decoupling at $p = 4$ for the cap pieces $f_\theta = \sum_{q \in Q_\theta} e(q \cdot x)$ gives $\|f_Q\|_{L^4} \lesssim_\epsilon \delta^{-\epsilon} (\sum_\theta \|f_\theta\|_{L^4}^2)^{1/2}$, and $\|f_\theta\|_{L^4}^4 = E_+(Q_\theta)$. Squaring twice yields the inequality. $\square$

Numerical consistency (dr2_decoupling_iteration.py, 3/3). The constant $K = E_+(Q)/(\sum_\theta E_+(Q_\theta)^{1/2})^2$ is ≤ 1.94 on great-circle, cap-grid and random sets at cap scales $n = 2, 4, 8$, and STABLE across scales (within $\times 1.25$): the iteration holds with a small, non-blowing-up constant, as the $\delta^{-\epsilon}$ factor predicts. The partition also breaks the additive structure ($\sum_\theta E_+(Q_\theta) \approx 220$ vs $E_+(Q) \approx 10^4$ at $n = 8$): the cross-cap quadruples dominate, and the iteration converts them into the controlled $(\sum \sqrt{E_\theta})^2$ form.

3. The induction-on-scales

Base case (separated). If Q is $\delta^{1/2}$ -separated, each cap holds ≤ 1 point, $E_+(Q_\theta) \leq 1$, so $\sum_\theta E_+(Q_\theta)^{1/2} \leq N$ and the Lemma gives $E_+(Q) \lesssim_\epsilon \delta^{-\epsilon} N^2$. With $\delta \sim (\text{minimal separation})^2 \geq N^{-O(1)}$, $\delta^{-\epsilon} \leq N^{O(\epsilon)}$, so $E_+(Q) \lesssim_\epsilon N^{2+\epsilon}$ – this is v1.1 §3.

Recursion (clustered). If some cap θ holds $m_\theta > 1$ points, $E_+(Q_\theta)$ is a SUB-problem on a $\delta^{1/2}$ -cap. By the affine-invariance rescaling (R-023), the parabolic map T_θ sends θ to a unit paraboloid patch with $E_+(Q_\theta) = E_+(T_\theta Q_\theta)$ – a FRESH instance of the same problem at unit scale with $m_\theta < N$ points. Apply the Lemma to each, recursing. The recursion terminates when every

active cap is separated (depth $O(\log(\text{dynamic range of separations})) = O(\log N)$ for sets with $\geq N^{-O(1)}$ separations; deeper otherwise).

The cited step. Substituting the inductive bound $E_+(Q_\theta) \lesssim_\epsilon m_\theta^{2+\epsilon}$ and summing, the naive accumulation gives $\delta^{-\epsilon}$ PER LEVEL. The standard Bourgain–Demeter / Bourgain–Guth multi-scale organisation (geometric choice of the per-level ϵ_k , or the fact that ℓ^2 -decoupling is ALREADY the multi-scale theorem and one application at the critical scale carries a single $\delta^{-\epsilon}$) keeps the TOTAL loss at $N^{O(\epsilon)}$, yielding $E_+(Q) \lesssim_\epsilon N^{2+\epsilon}$ for arbitrary finite Q . *This bookkeeping is cited, not reproduced here.*

4. Status

The cross-scale energy summation (the operator’s named residual) is now WRITTEN as the decoupling-iteration inequality (§2, derived conditional on decoupling and numerically consistent) plus the affine-invariance recursion (§3). The single remaining cited-not-reproduced ingredient is the tight per-level constant control. The unrestricted DR-2 is therefore lifted from T4 to T5 (a structurally-complete reduction). It is NOT promoted to T6 here, and NO gate/tier flip is made: whether the cited bookkeeping is accepted as textbook-standard (which would make DR-2 = T6 PROVED CONDITIONAL on decoupling and permit a DR2-SHARE flip + removing H-ADM-COH from B1) is an operator decision. DR2-SHARE stays OPEN; B5 T5; B1 T6; H-ADM-COH retained.

5. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3, code-discipline 6)

- (α) “*The iteration inequality is conditional on decoupling.*” DECLARED: yes; §2 derives it FROM decoupling at $p = 4$, it is not independent. The note never claims to prove decoupling.
- (β) “*The per-level constant control is not written.*” UPHELD and the reason the grade is T5, not T6: the geometric ϵ -bookkeeping over $O(\log N)$ levels is cited (Bourgain–Demeter), not reproduced. The iteration constant is numerically bounded ($K \leq 1.94$, scale-stable), which is consistent with but not a proof of the rigorous control.
- (γ) “*Does very small minimal separation break it?*” ADDRESSED: when points are closer than $N^{-O(1)}$ the single-scale bound is loose, which is exactly why the recursion (R-023 rescaling) is needed; rescaling resets the scale so the loss is per-level $O(1)$, not tied to the absolute minimal separation. The numerics (K scale-stable) support this; the rigorous summation is the cited step.

Quantitative sanity check (mandatory): magnitude – $K \leq 1.94$ (small), and $\sum_\theta E_+(Q_\theta) \approx 2 \times 10^2 \ll E_+(Q) \approx 10^4$ (the iteration genuinely reduces); sign-direction – the inequality is an upper bound, all terms positive; limit-case – separated Q gives the v1.1 §3 base case exactly; consistency – K stable across cap scales ($\times 1.25$) matches the $\delta^{-\epsilon}$ prediction of a mild per-level constant.

6. Result footer

Result ID:	B5 / DR-2 cross-scale induction (decoupling-iteration)
Precise statement:	Conditional on l2-decoupling at p=4, the iteration $E_+(Q) \leq \epsilon \sum_\theta \sqrt{E_+(Q_\theta)}$ holds over $\delta^{1/2}$ -caps (derived from the v1.1

majorant); with the R-023 rescaling recursion it reduces arbitrary finite Q to the separated base case ($\rightarrow N^{\{2+\epsilon\}}$). The tight per-level constant bookkeeping ($O(\log N)$ levels $\rightarrow$ total $N^{\{0(\epsilon)\}}$) is cited, not reproduced. Iteration constant numerically bounded ($K \leq 1.94$, scale-stable).

Scope: Unrestricted-class sphere additive energy; the cross-scale step of DR-2 / DR2-SHARE.

Dependencies: 12-decoupling (BLACK BOX); v1.1 Section 2.1 (majorant); R-023 (affine-invariance rescaling); standard multi-scale bookkeeping (CITED).

Evidence grade: DERIVED (iteration inequality, conditional on decoupling) + EXECUTED numerics (dr2_decoupling_iteration.py 3/3) + CITED (per-level constant control).

Reproduction command: python codes/vacuum/dr2_decoupling_iteration.py

Expected output: 3/3 PASS; $K \leq 1.94$ across configs/scales; partition reduces additive structure.

Falsification gate: A finite Q with $K = E_+(Q)/(\sum \sqrt{E_{\text{theta}}})^2$ growing polynomially in N (would contradict the decoupling exponent); not observed.

Tier before / after: DR-2 unrestricted: T4 STRONG EVIDENCE $\rightarrow$ T5 (structurally complete reduction). NO flip. Separated case stays T6 conditional. B5 T5, B1 T6, H-ADM-COH retained, DR2-SHARE OPEN.

No-overclaim statement: NOT a DR-2 closure and NOT T6. The cross-scale step is written and numerically consistent, but the tight per-level constant control is cited, not reproduced. Decoupling is invoked, not reproved. No gate/tier/hypothesis action.

Next required action: Either reproduce the per-level constant bookkeeping in full ($\rightarrow$ T6 PROVED CONDITIONAL, operator may then flip DR2-SHARE), or accept the cited standard machinery by operator sign-off.